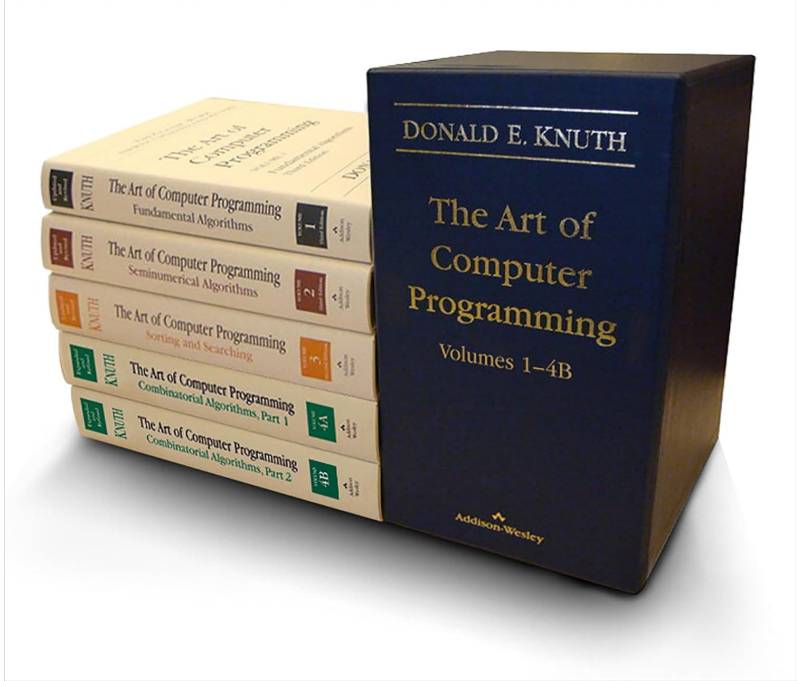


Donald E. Knuth



dobitnik Turingove nagrade (1974)



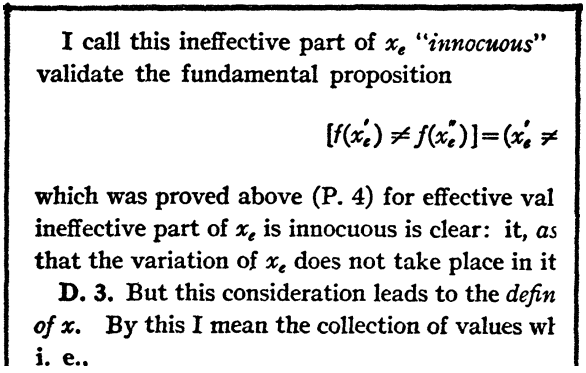

DONALD E. KNUTH
The Art of Computer Programming
Volumes 1–4B
Addison-Wesley
The Art of Computer Programming
Fundamental Algorithms
The Art of Computer Programming
Seminumerical Algorithms
The Art of Computer Programming
Sorting and Searching
The Art of Computer Programming
Combinatorial Algorithms, Part 1
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Combinatorial Algorithms, Part 2
KNUTH

- Vol. 1 (1968)
- Vol. 2 (1969)
- Vol. 3 (1973)

- Vol. 4A (2011)
- Vol. 4B (2022)

- Vol. 1 (1968)
- Vol. 2 (1969)
- Vol. 3 (1973)
- Vol. 2, druga izdaja (1976–1981)
- Vol. 4A (2011)
- Vol. 4B (2022)

Transactions of the American Mathematical Society



I call this ineffective part of x_e "*innocuous*" validate the fundamental proposition

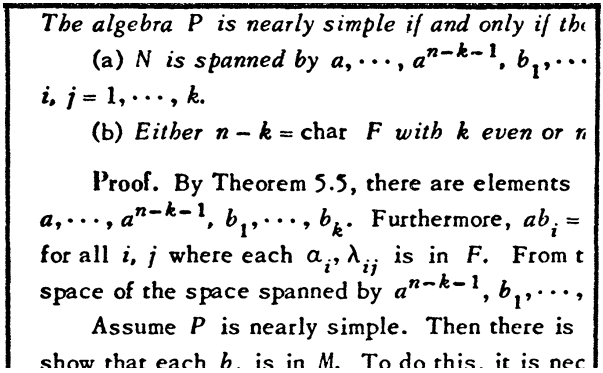
$$[f(x_e') \neq f(x_e'')] = (x_e' \neq$$

which was proved above (P. 4) for effective val ineffective part of x_e is innocuous is clear: it, *as* that the variation of x_e does not take place in it

D. 3. But this consideration leads to the *defin of* x. By this I mean the collection of values wh i. e..

zvezek 23 (1922)

D. E. Knuth, *Mathematical Typography*, Bull. AMS 1 (1979), 337–372



The algebra P is nearly simple if and only if th

(a) *N is spanned by $a, \cdots, a^{n-k-1}, b_1, \cdots$ $i, j = 1, \cdots, k$.*

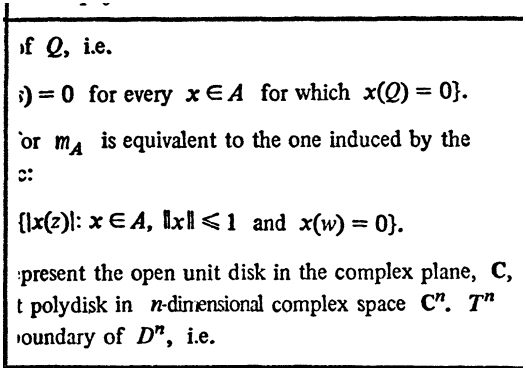
(b) *Either $n - k =$* char *F with k even or n*

Proof. By Theorem 5.5, there are elements $a, \cdots, a^{n-k-1}, b_1, \cdots, b_k$. Furthermore, $ab_i =$ for all i, j where each α_i, λ_{ij} is in F. From t space of the space spanned by $a^{n-k-1}, b_1, \cdots,$

Assume P is nearly simple. Then there is show that each b_i is in M. To do this, it is nec

zvezek 169 (1972)

D. E. Knuth, *Mathematical Typography*, Bull. AMS 1 (1979), 337–372



of Q, i.e.

$\}) = 0$ for every $x \in A$ for which $x(Q) = 0\}$.

or m_A is equivalent to the one induced by the

:

$\{|x(z)|: x \in A, \|x\| \leqslant 1$ and $x(w) = 0\}$.

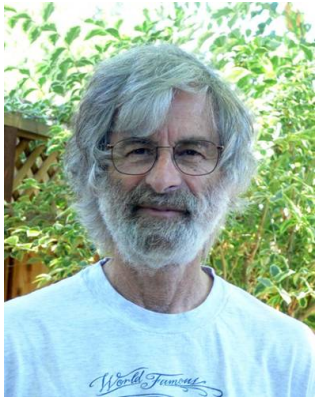
present the open unit disk in the complex plane, $\mathbf{C}$,
t polydisk in n-dimensional complex space $\mathbf{C}^n$. T^n
oundary of D^n, i.e.

zvezek 199 (1974)

D. E. Knuth, *Mathematical Typography*, Bull. AMS 1 (1979), 337–372

TEX (1978)

Leslie Lamport



dobitnik Turingove nagrade (2013)

LaTeX (1984)

2026?

Markdown

```
# Naslov

Besedilo je lahko **poudarjeno** ter
vsebuje [povezave](https://www.fmf.uni-lj.si/),
`programsko kodo` in sezname:
- prva točka
- druga točka
- tretja točka
```

HTML

```
<h1>Naslov</h1>

<p>Besedilo je lahko <strong>poudarjeno</strong> ter
vsebuje <a href="https://www.fmf.uni-lj.si/">povezave</a>,
<code>programsko kodo</code> in sezname:</p>
<ul>
  <li>prva točka</li>
  <li>druga točka</li>
  <li>tretja točka</li>
</ul>
```

```
\section{Naslov}

Besedilo je lahko \textbf{poudarjeno} ter
vsebuje \href{https://www.fmf.uni-lj.si/}{povezave},
\texttt{programsko kodo} in sezname:

\begin{itemize}
  \item prva točka
  \item druga točka
  \item tretja točka
\end{itemize}
```

Ukazi in okolja

Ukazi

- `\ukaz`
- `\ukaz{argument}`
- `\ukaz[neobvezen]{argument}`
- `\ukaz[neobvezen]{argument1}{argument2}`

Okolja

```
\begin{okolje}
  ...
\end{okolje}
```

Slabo

```
Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Ut id viverra ligula. Phasellus vehicula lorem vitae luctus
dignissim.
Sed ac justo commodo, fringilla urna ac, efficitur leo.
Praesent dui odio, accumsan ac sapien nec, interdum volutpat
est.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut id viverra ligula. Phasellus vehicula lorem vitae luctus dignissim. Sed ac justo commodo, fringilla urna ac, efficitur leo. Praesent dui odio, accumsan ac sapien nec, interdum volutpat est.

Nove odstavke naredimo s prazno vrstico

Dobro

```
Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Ut id viverra ligula. Phasellus vehicula lorem vitae luctus
dignissim.

Sed ac justo commodo, fringilla urna ac, efficitur leo.
Praesent dui odio, accumsan ac sapien nec, interdum volutpat
est.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut id viverra ligula. Phasellus vehicula lorem vitae luctus dignissim.

Sed ac justo commodo, fringilla urna ac, efficitur leo. Praesent dui odio, accumsan ac sapien nec, interdum volutpat est.

Novih odstavkov se ne dela z \\

Sprejemljivo

```
Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Ut id viverra ligula. Phasellus vehicula lorem vitae luctus
dignissim. \par
Sed ac justo commodo, fringilla urna ac, efficitur leo.
Praesent dui odio, accumsan ac sapien nec, interdum volutpat
est.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut id viverra ligula. Phasellus vehicula lorem vitae luctus dignissim.

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Slabo

```
Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Ut id viverra ligula. Phasellus vehicula lorem vitae luctus
dignissim. \\
Sed ac justo commodo, fringilla urna ac, efficitur leo.
Praesent dui odio, accumsan ac sapien nec, interdum volutpat
est.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut id viverra ligula. Phasellus vehicula lorem vitae luctus dignissim.
Sed ac justo commodo, fringilla urna ac, efficitur leo. Praesent dui odio, accumsan ac sapien nec, interdum volutpat est.

Novih odstavkov se ne dela z \\

Slabo

```
Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Ut id viverra ligula. Phasellus vehicula lorem vitae luctus
dignissim. \\[1ex]
Sed ac justo commodo, fringilla urna ac, efficitur leo.
Praesent dui odio, accumsan ac sapien nec, interdum volutpat
est.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut id viverra ligula. Phasellus vehicula lorem vitae luctus dignissim.

Sed ac justo commodo, fringilla urna ac, efficitur leo. Praesent dui odio, accumsan ac sapien nec, interdum volutpat est.

Struktura dokumenta

```
\documentclass[11pt,a4paper]{article}
% preambula
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage[slovene]{babel}

\title{Naslov dokumenta}
\author{Avtor dokumenta}
\date{\today}

\begin{document}
\maketitle
...vsebina...
\end{document}
```

Ne vklapljajte nepotrebnih paketov

Dobro

```
\documentclass[11pt]{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}

\usepackage[slovene]{babel}
\usepackage{lmodern}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amsthm}
\usepackage{mathtools}
```

Slabo

```
\documentclass[11pt]{article}

\usepackage[utf8]{inputenc}
% Ostale možnosti za kodiranje so:
% Kodirna tabela za Windows
% \usepackage[cp1250]{inputenc}
% Kodirna tabela za Linux
% \usepackage[latin2]{inputenc}
\usepackage[T1]{fontenc}

\usepackage[slovene]{babel}
\usepackage{lmodern}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amsthm}
\usepackage{mathtools}
\usepackage{graphicx}
% \usepackage{tikz}
```

Pomožne datoteke

```
dokument.aux
dokument.fdb_latexmk
dokument.fls
dokument.log
dokument.pdf
dokument.synctex.gz
dokument.tex
```

.gitignore

```
*.aux
*.fdb_latexmk
*.fls
*.log
*.synctex.gz
*.pdf
```

Datoteka `.gitignore` velja samo za datoteke, ki jim Git še ne sledi.

Matematika

$$e^{i\pi} + 1 = 0$$

```
e^{i \pi} + 1 = 0
```

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

```
\lim_{x \to 0} \frac{\sin x}{x} = 1
```

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

```
\int_{-\infty}^\infty e^{-x^2}\,dx = \sqrt{\pi}
```

LaTeX izven LaTeXa

Uporabiti morate ustrezne ukaze

Slabo

```
x \epsilon \mathbb{R}
```

$$x\epsilon\mathbb{R}$$

Dobro

```
x \in \mathbb{R}
```

$$x \in \mathbb{R}$$

Uporabiti morate ustrezne ukaze

Slabo

```
\Sigma_{i=1}^n a_i^2 = 42
```

$$\Sigma_{i=1}^n a_i^2 = 42$$

Dobro

```
\sum_{i=1}^n a_i^2 = 42
```

$$\sum_{i=1}^{n} a_i^2 = 42$$

Uporabiti morate ustrezne ukaze

Slabo

```
A \cap B = \phi
```

$$A \cap B = \phi$$

Dobro

```
A \cap B = \emptyset
```

$$A \cap B = \emptyset$$

Uporabiti morate ustrezne ukaze

Slabo

```
2 < x, y > + 3 < y, x >
```

$$2 < x, y > +3 < y, x >$$

Dobro

```
2 \langle x, y \rangle
+ 3 \langle y, x \rangle
```

$$2\langle x, y\rangle + 3\langle y, x\rangle$$

Za matematične operacije uporabimo ustrezne ukaze

Slabo

```
sin x + sin (x + y)
```

$$sinx + sin(x + y)$$

Slabo

```
sin~x + sin~(x + y)
```

$$sin\ x + sin\ (x + y)$$

Slabo

```
\mathsf{sin}~x + \mathsf{sin}~(x + y)
```

$$\mathsf{sin}\ x + \mathsf{sin}\ (x + y)$$

Dobro

```
\sin x + \sin (x + y)
```

$$\sin x + \sin(x + y)$$

Vrstični in prikazni način

Naj bodo $a, b, c \in \mathbb{R}$. Tedaj je

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

rešitev enačbe $ax^2 + bx + c = 0$.

```
Naj bodo \(a, b, c \in \mathbb{R}\). Tedaj je
\[
  x_{1, 2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
\]
rešitev enačbe \(ax^2 + bx + c = 0\).
```

Vrstični in prikazni način

Naj bodo $a, b, c \in \mathbb{R}$. Tedaj je

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

rešitev enačbe $ax^2 + bx + c = 0$.

```
Naj bodo $a, b, c \in \mathbb{R}$. Tedaj je
$$
  x_{1, 2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
$$
rešitev enačbe $ax^2 + bx + c = 0$.
```

Za sredinsko poravnane izraze uporabljamo prikazni način

Slabo

```
\begin{center}
  \( e^{i \pi} + 1 = 0 \)
\end{center}
```

$$e^{i\pi} + 1 = 0$$

Dobro

```
\[
  e^{i \pi} + 1 = 0
\]
```

$$e^{i\pi} + 1 = 0$$

Za sredinsko poravnane izraze uporabljamo prikazni način

Slabo

```
\begin{center}
  \( x_{1, 2} = ... \)
\end{center}
```

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Dobro

```
\[
  x_{1, 2} = ...
\]
```

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$